

Introduction

This lab consisted of using a water level control system and a motor control system.

The water level control system contained a reservoir that would hold a large amount of water, and another system that would fill a smaller tank. The smaller tank had a floating mass connected to a weight on a pulley and would provide an output voltage based on the position of the floating mass and weight. This output voltage is what is recorded in my analysis. The output voltage and input voltage were able to be recorded and set by the PC by using a Data Acquisition Device (DAQ).

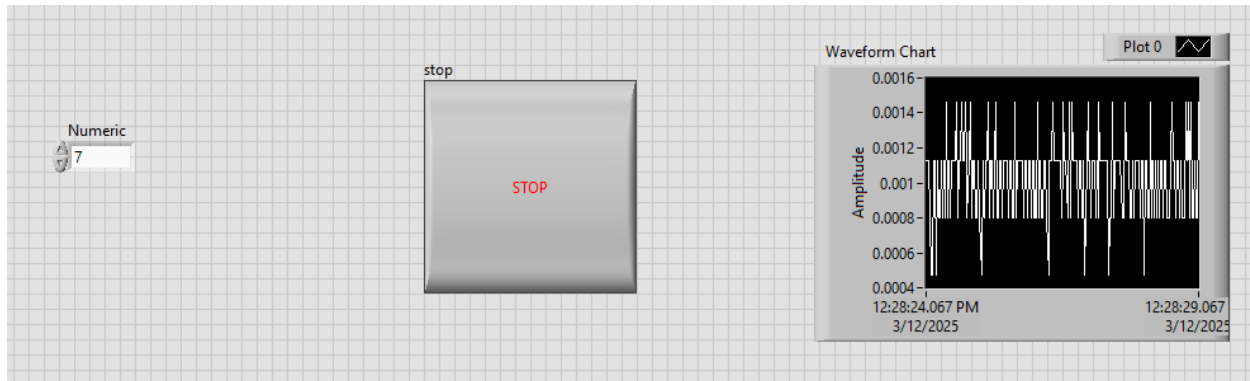
Notably for the water control system there was an external power supply that turned on the motors and sensors running the hardware, while the PC only controlled stuff using the DAQ.

The motor control system was a bit simpler and had a servo motor connected to the PC using an interface cable. The motor would change speed based on the input voltage set by the user. This speed was measured using a handheld tachometer to measure the speed in revolutions per minute (RPM).

Analysis

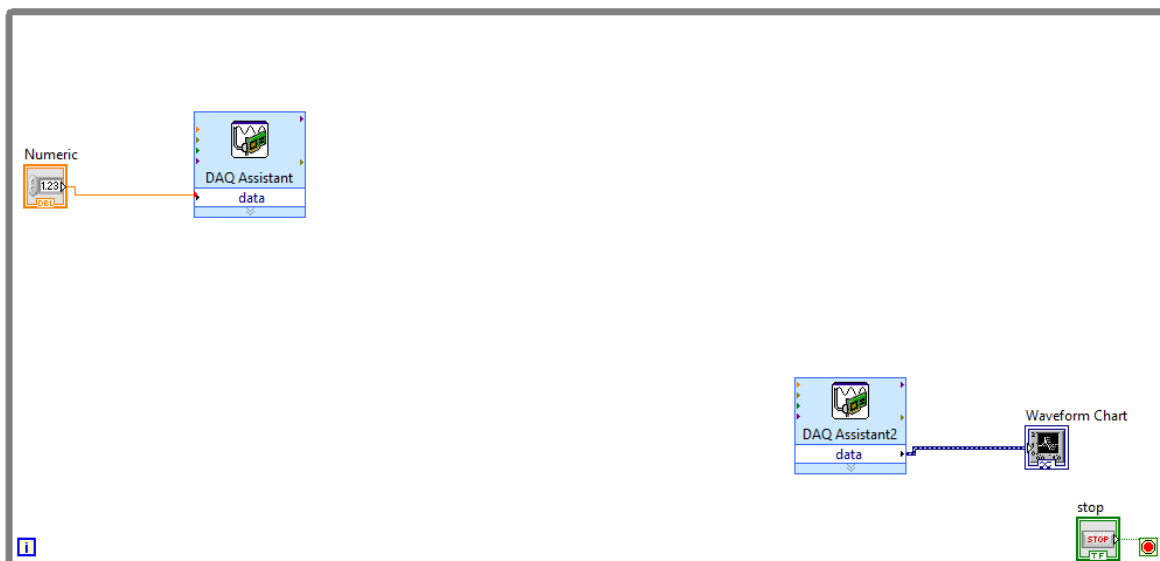
Lab 7a was the water control part of the lab.

Lab 7a Front Panel:



This was a simple program. It contains a numeric input for the input voltage and a waveform chart to display the output voltage. There is a while loop connected to a stop button whenever the user wants to stop the program.

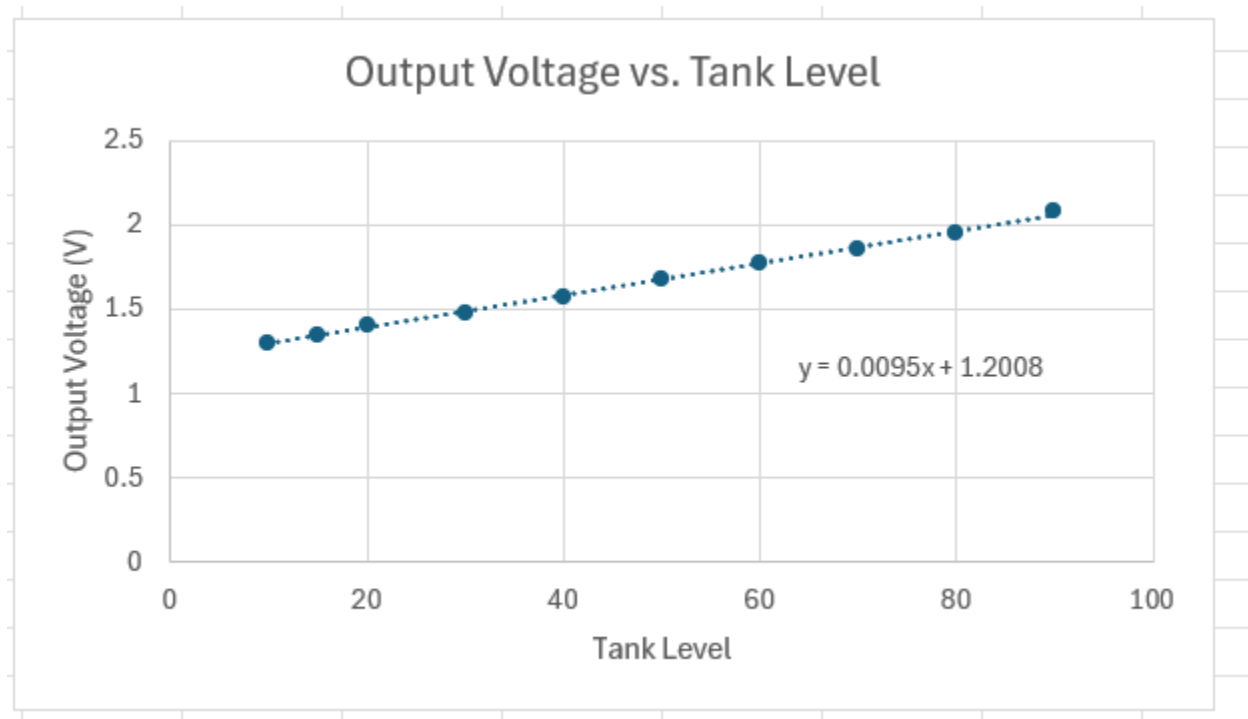
Lab 7a Block Diagram:



The block diagram contains a while loop to run as long as the user desired. The DAQ generates data from the numeric control as an input, and the DAQ will acquire data as an output and display that to the waveform chart. Both DAQs were configured to an A0 input/output. Notably the generating DAQ had to be restricted to have a minimum value of 0 V to prevent the motor from rotating backwards and damaging the gears. Likewise, the DAQ was configured to measure an infinite number of values (as according to the while).

For the first part of the experiment, I was tasked with recording the output voltage of the sensor based on the current tank level. The input voltage was not that important, so I kept it around 1 V. I would attempt to stabilize the tank at a certain level by letting it fill with water a while and control the outflow of the tank's water.

Float level Transmitter			
Input Voltage	Tank Level	Output Voltage	
1 V	10	1.3	
	15	1.35	
	20	1.4	
	30	1.475	
	40	1.575	
	50	1.673	
	60	1.77	
	70	1.857	
	80	1.95	
	90	2.08	maxes out the voltage at 90



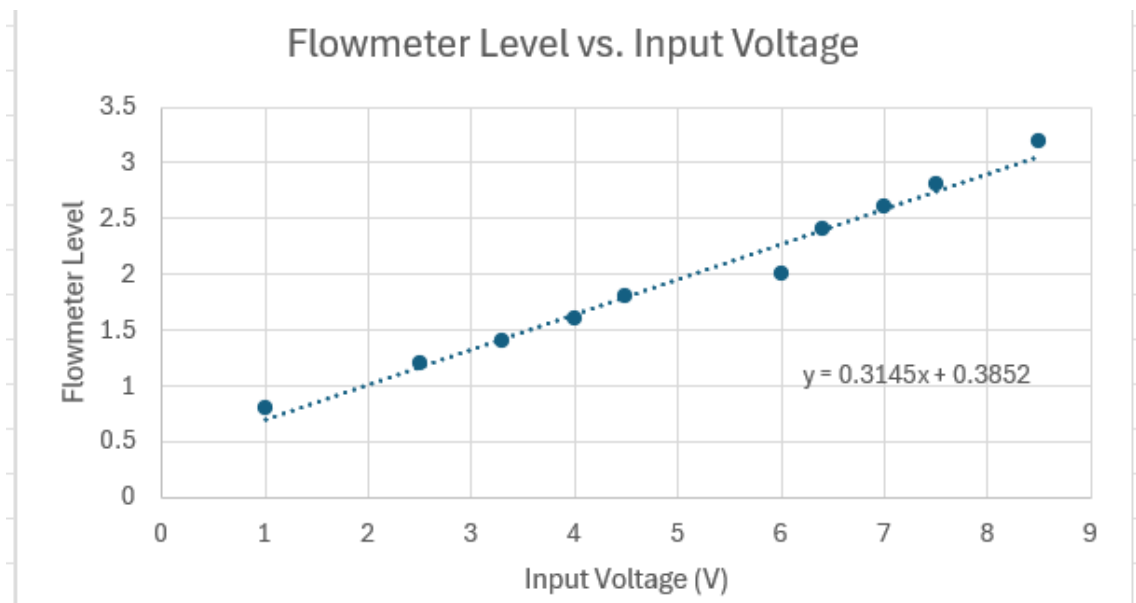
The transfer function of a linear system of data is the slope of the line, so

$$G(s) = 0.0095$$

is the transfer function of this system.

For the second part of this experiment, I was tasked with seeing the effect of the input voltage on the flowmeter. The output voltage was irrelevant for this part, and remained around a constant of 0.4 V. This one was a bit easier to measure, as anytime I changed the input voltage the flowmeter would stay at a constant level until I changed the input voltage again.

Pulse Flow Transmitter	
Input Voltage (V)	Flowmeter Level
1	0.8
2.5	1.2
3.3	1.4
4	1.6
4.5	1.8
6	2
6.4	2.4
7	2.6
7.5	2.8
8.5	3.2



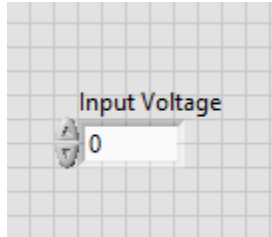
The transfer function of a linear system of data is the slope of the line, so

$$G(s) = 0.3145$$

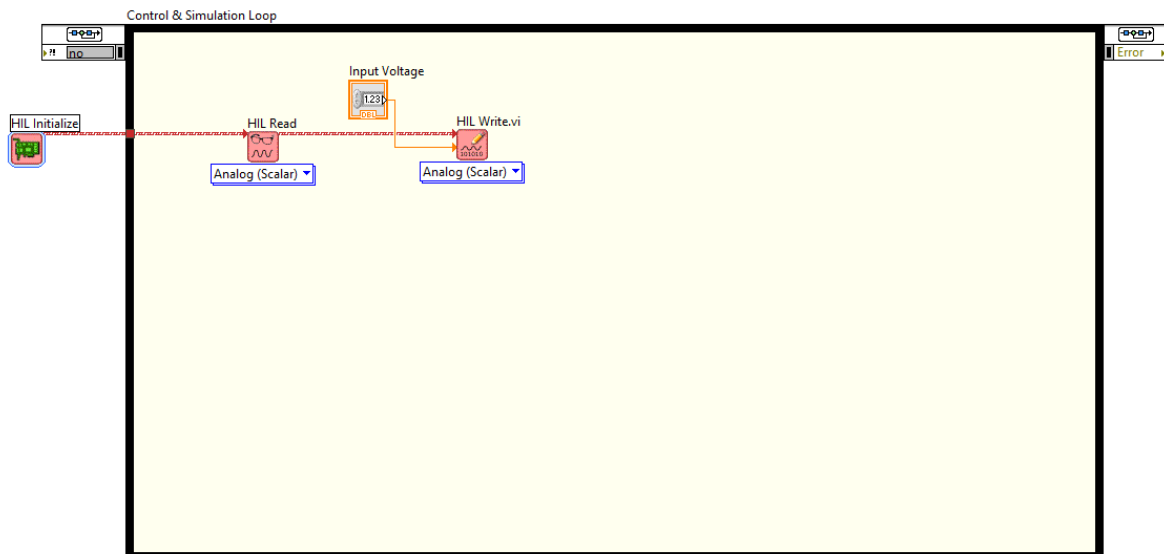
is the transfer function of this system.

Lab 7b was the motor control part of the lab.

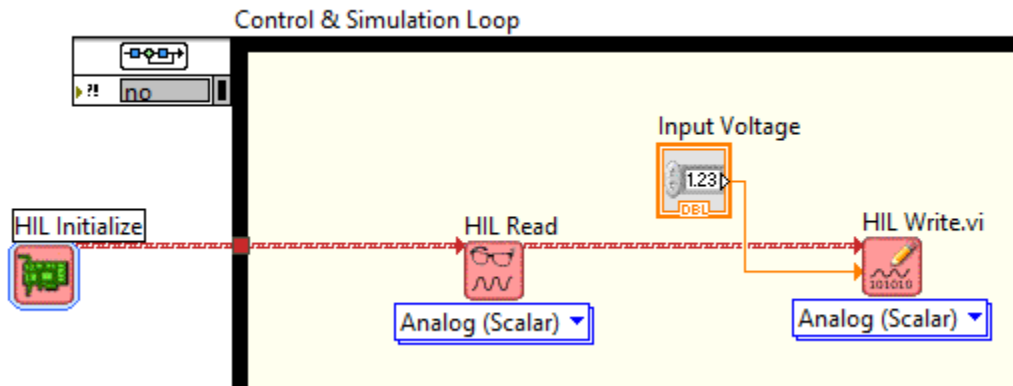
Lab 7b Front Panel:



Lab 7b Block Diagram:



Zoomed in view of the important stuff in Block Diagram:

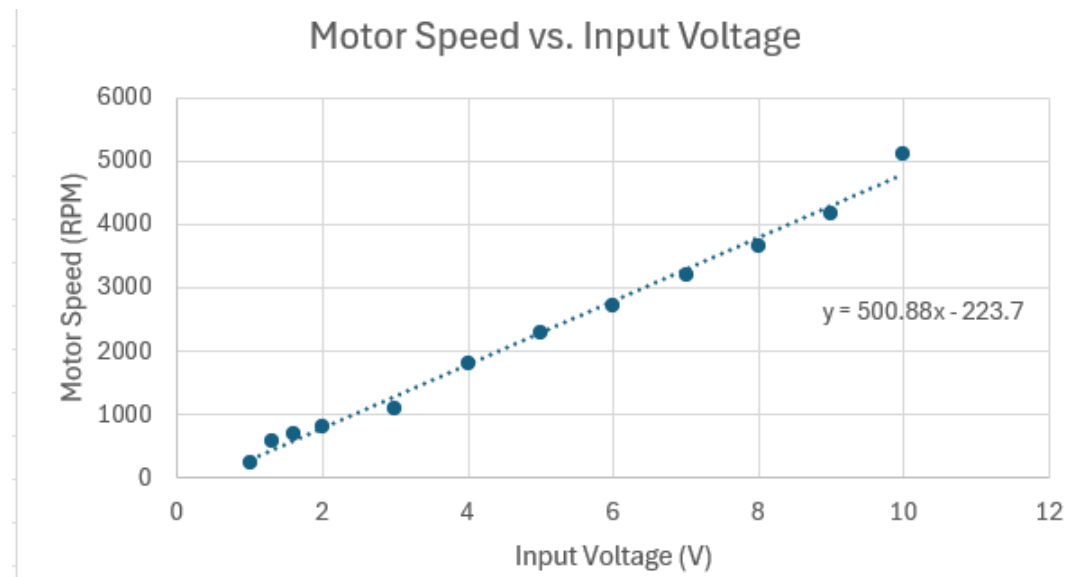


This was a lot of new stuff we did not spend any time going over in lecture. I was given and followed a step-by-step instruction guide to create this block diagram.

These items were selected by going down the menu in the block diagram and following the instructions. I connected them following the given pictures (notably input voltage was connected to HIL Write's "voltage" input).

I adjusted the input voltage using the numeric control on the front panel, and I measured the motor speed in RPM using a handheld tachometer. A piece of tape was placed on the motor, and the tachometer would attempt to measure the RPM using that tape. I would have to wait for the tachometer to stabilize, like how I waited for the output voltage to stabilize with the water control system.

Input Voltage (V)	Motor Speed (RPM)
1	220.5
1.3	575
1.6	700
2	800.2
3	1100
4	1800
5	2300
6	2700
7	3200
8	3661
9	4160
10	5100



The transfer function of a linear system of data is the slope of the line, so using Excel's trendline feature:

$$G(s) = 500.88$$

is the transfer function of this system.

Conclusion

This lab was odd. It was interesting to measure actual physical systems and see how the hardware interfaces with the PC. Likewise, it was interesting to generate a data file myself to incrementally step through the labs and verify/discover that the motors follow a linear relationship with the voltage. This is most easily seen by the motor control system as the motor speed is directly measured and proportional to the voltage.

This will be useful in the future for adapting to using other hardware and learning new control systems as needed.